

Understanding LLM Usage Behavior of Online Survey Participants

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*Describe a time you noticed other people handling challenges at work. What was challenging about what they did?

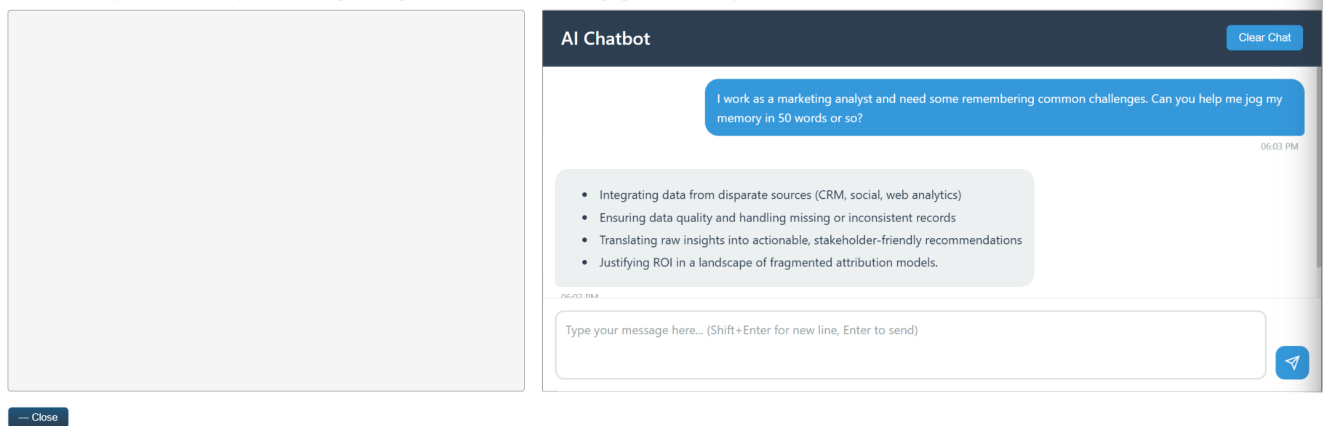


Figure 1: We built a custom survey interface where participants can optionally use an AI chatbot we provided to help them answer their questions. We captured how participants prompted the chatbot and how they incorporated the output in their question responses.

Abstract

As large language models (LLMs) become widely accessible, online study participants increasingly use LLM tools such as ChatGPT when completing surveys, especially for open-ended questions. Such use raises concerns about response quality, as LLM-generated text may not accurately reflect participants' actual lived experiences and opinions. In this pilot study, we capture how participants use AI chatbots when answering survey questions by instrumenting an online survey with a custom AI chatbot side-by-side with survey questions. We investigate how different question types, as well as the effort and personalness of questions, impact participants' propensity to use the AI chatbot. Preliminary findings suggest that when the AI chatbot is used, participants use initial prompts very similar to the question text and their question responses are very similar to the LLM output. Higher perceived effort is associated with

increased odds of LLM use, and anecdotal questions show lower overlap with LLM-generated content than generative questions.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI; User studies.*

Keywords

data quality, online surveys, study participants, LLM, chatbot

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1 Introduction

Open-ended, unmoderated survey questions are foundational methods to elicit subjective, human experiences from participants. However, the rapid adoption of large language models (LLMs) has begun



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to challenge a core assumption underlying these methods: that open-ended responses are authored directly by participants. Recent studies report that a substantial portion of online participants use tools such as ChatGPT when completing surveys, especially for open-ended questions [20, 21]. Participants describe using LLMs to draft, expand, clarify, or polish responses, and in some cases, to generate answers entirely [13]. As a result, responses may appear fluent and thoughtful but may no longer faithfully reflect participants' own experiences.

Prior work shows that LLM-generated or LLM-assisted text tends to be more homogeneous than human-authored responses [19, 21]. At the same time, traditional data-quality safeguards—such as attention checks, response length thresholds, or timing-based filters—are often ineffective at detecting such use [16, 20]. Rather than producing low-effort or nonsensical answers, participants may now offload cognitive and expressive labor to LLMs. This shift is particularly concerning for data quality in HCI research.

Insight into participants' LLM use is currently based on participants' self-reported usage [13, 21] and measured at the participant level. However, we lack insight into in situ LLM behaviors and variances in usage throughout a survey, from question to question. Little is known about the role of question characteristics, such as personalness, cognitive load, or function (e.g., anecdotal vs. generative), on participants' decision to use an LLM and the extent to which LLM-collaboration influences the final response.

In this pilot study, we explore these gaps through an empirical examination of LLM-assisted responding in online surveys. Rather than relying solely on self-reports, we instrument survey participation with a custom AI chatbot to directly measure when and how LLMs are used across predefined question types. Focusing on open-ended questions that elicit experiential and opinion-based responses—common and especially vulnerable in HCI research—we make two primary contributions:

- **Fine-grained measurement of LLM involvement.** We measure the proportion of LLM-authored content and the role LLM plays in response production in situ.
- **A mapping between question characteristics and LLM involvement.** We provide preliminary evidence on how question properties including personalness, cognitive effort, and question type (anecdotal vs. generative) may influence whether and how participants use LLMs.

2 Related Work

Recent work has documented substantial [21] and non-uniform use of LLMs among participants in online research. In a representative Prolific survey, 34% of participants reported at least occasional LLM use (with 9% reporting frequent use), while task-based studies estimate prevalence between 33–46% for text-production tasks [20].

Other work has examined how LLM use manifests in qualitative and interactive research contexts. In online focus groups, over 20% of chat responses from participants opting to respond via chat rather than speech closely resembled ChatGPT outputs [17]. These responses were often marked by abrupt shifts to formal tone, American spelling in UK-based samples, and generalized or cautionary language. Similarly, experienced participants use LLMs not only to

draft responses, but also to clarify confusing instructions and “polish” rough ideas into the detailed narratives they believe researchers expect [13].

A large body of prior work in survey methodology shows that question characteristics strongly shape response behavior. Questions that impose high cognitive effort frequently elicit satisficing, where respondents moderate their effort by providing superficial or approximate answers rather than fully engaging with the task [1, 4, 7–9, 14].

Beyond effort, question type also shapes the cognitive processes involved in responding, with anecdotal questions requiring lower-order recall processes and generative questions requiring higher-order processes like synthesis and creation [3, 6, 15]. Higher-order questions typically induce greater cognitive load [7, 18], while lower-level questions, although easier to answer, remain vulnerable to memory failures and reporting error.

Personalness and self-disclosure further influence response production. When questions involve sensitive or identity-relevant topics, respondents often adopt strategies to manage privacy risks, including socially desirable responding, inaccurate or fabricated answers, item skipping, or deliberate blurring of details [2, 5, 12]. Together, these findings suggest that effort, question type, and personalness systematically shape how responses are produced, particularly in open-ended formats that rely on articulation and reflection.

Despite these insights, prior work has not systematically examined whether specific question characteristics—such as cognitive effort, question type, and personalness—predict participants' propensity to delegate response production to LLMs during survey participation. Addressing this gap, the present study investigates how these factors jointly shape LLM use in open-ended survey responses and the resulting composition of LLM–human hybrid data by answering the following research questions:

- **RQ1:** How do participants use AI chatbots when completing unmoderated, open-ended online surveys?
- **RQ2:** How does AI chatbots use vary with question characteristics (personalness and cognitive effort) and question type (anecdotal vs. generative)?

3 Methods

3.1 Study Design

We designed an online study where participants answered open-ended questions and could optionally use a provided AI chatbot per question (see Figure 1). First, we developed a set of questions on the broadly accessible topic of careers for two question types: anecdotal (e.g., “Describe a time you had to convince others to support your idea. Looking back, what would you repeat or do differently next time?”) and generative (e.g., “Imagine you had to start over in a completely new field, describe how might you decide where to begin?”). Each participant then answered nine questions from one question type in randomized order; the set of nine questions varied in level of personalness and cognitive effort. Participants could optionally click a button “Open the LLM Chatbot” to reveal our provided AI chatbot next to the question and use it in any way they wanted to help them answer the question. The chatbot remained hidden until

the button was affirmatively clicked to avoid inducing use. Following each question, participants evaluated the perceived effort and personalness using a 7-point scale (1 = not at all effortful/personal, 7 = extremely effortful/personal). Participants were informed that their compensation would not be affected by whether or not they used the chatbot. Surveys ended with questions regarding their use of the provided AI chatbot as well as basic demographic questions.

3.1.1 Embedded LLM Chatbot. We embedded a custom-built AI chatbot webapp (powered by OpenLLM’s open source gpt-oss-120b model [10]) into a Qualtrics survey to be used side-by-side with survey questions (see Figure 1). We provided system prompts to keep the response brief and avoid tables and other complex formatting. Each survey question had a blank chatbot (history was not retained between questions), and participants were given a sample question at the start to familiarize themselves with the tool. Chatbot prompts and outputs were logged for analysis.

3.2 Participants

20 participants (9 female; 11 male; average age of 44) were recruited through Prolific [11] to participate in a survey about career opinions. Ten participants answered anecdotal questions and ten participants answered generative questions. Participants were at least 18 years of age or older, had a minimum 95% approval rating and 100 submissions on the platform, were either employed part-time or full-time, interacted with an AI chatbot at least once a week, and were at least moderately confident in using an AI chatbot. Participants were compensated at the rate of \$17/hr and took on average 37.4 minutes to complete the study.

4 Preliminary Results

4.1 Overview of LLM Use Behavior (RQ1)

Overall, 9 of 20 participants used the chatbot at least once (excluding the practice question), defined as any message sent to the AI chatbot regardless of length or whether generated content was incorporated into the final response. Across chatbot-assisted responses, participants engaged in relatively brief interactions, averaging 3.09 turns per interaction ($SD = 1.81$, $range = 2.0-8.0$). Participant messages averaged 25 words ($SD = 15.45$) and 138 characters ($SD = 90$).

4.1.1 How Participants Prompted the Chatbot. We examined how closely participants’ initial prompts aligned with the study questions. Cosine similarity was computed between the question text and initial prompt to the assistant using TF-IDF character n-gram features (2–3 grams). Lexical overlap was measured as the percentage of unique response word tokens that appeared in both the question text and initial prompt, after regex-based tokenization. Across all LLM-assisted responses ($N = 33$), participants’ initial prompts showed relatively high cosine similarity and lexical overlap with the original question text (cosine similarity: $M = 0.73$, $SD = 0.34$; lexical overlap: $M = 0.70$, $SD = 0.35$). Prompt-token coverage was also high on average ($M = 0.86$, $SD = 0.30$), suggesting that participants largely reused the original prompt questions when first interacting with the chatbot, with minimal additional elaboration.

Whereas 5 of 9 participants directly copied at least one question prompt without any personal edits, others’ initial prompts

indicated the direction they wanted in their final response. For example, when the question asked broadly about career goals, P8 prompted the chatbot to “[D]escribe what it would be like to be a software engineer.” P1 and P4’s initial prompts further reveal how memory lapses may occur during open-ended questions and how participants try to collaborate with the chatbot to recover from such lapses: “I need to describe a time I faced a difficult task. I know we don’t know each other, but could you help me to jog my memory for something?” (P4).

4.1.2 How Participants Incorporated LLM-Generated Content into their Responses. Cosine similarity values between the chatbot’s content and participants’ submitted responses were generally high ($M \approx .79$, $SD \approx .20$). Notably, 50% of responses exhibited cosine similarity values above 0.88, and 75% exceeded 0.96, indicating substantial semantic overlap between LLM-generated content and participants’ final responses. At the same time, variability in similarity scores ($range = 0.23-1.00$) suggests heterogeneity in how participants integrated LLM-generated content, with some responses showing substantially transforming LLM-generated material.

4.2 LLM Use Varied by Question Characteristics (RQ2)

4.2.1 Effects of Perceived Effort and Personalness. We tested whether self-reported ratings of question characteristics predicted LLM use using separate logistic generalized estimating equation (GEE) models with participant as the clustering variable. Higher question effort ratings were associated with increased odds of LLM use ($\beta = .24$, $SE = .19$; $OR = 1.27$), but this effect was not statistically significant ($p = .19$). While the estimated effect of effort was positive, the small sample ($N = 20$) limited statistical power. Question personalness ratings did not predict LLM use ($\beta = .06$, $SE = .16$; $OR = 1.06$, $p = .70$).

4.2.2 Differences Between Anecdotal and Generative Question Types. Observed LLM usage was comparable across question types (5/10 participants for anecdotal, 4/10 participants for generative), but use intensity (messages sent per question) per user was higher overall for generative ($M = 1.95$) than for anecdotal ($M = 1.00$).

Initial prompts for anecdotal questions had high cosine similarity ($M = 0.79$), lexical overlap ($M = 0.76$), and prompt coverage ($M = 0.94$), indicating that participants often passed the prompt directly to the chatbot with minimal reinterpretation. In contrast, generative questions showed lower similarity across all metrics (cosine similarity: $M = 0.69$; lexical overlap: $M = 0.66$; prompt coverage: $M = 0.80$), suggesting that participants were more likely to actively reframe or elaborate the prompt for this question type.

Among responses that involved LLM usage, final responses to generative questions exhibited higher cosine similarity and lexical overlap with LLM output than responses to anecdotal questions. Using a conservative definition of near-verbatim reuse (≥ 0.80 cosine similarity and $\geq 90\%$ lexical overlap), approximately 79% (15 of 19) of generative responses met this criterion, compared to 21% (3 of 14) of anecdotal responses. Formal statistical tests did not reach conventional significance thresholds. However, effect sizes were sizable ($r \approx 0.3$ and $r \approx 0.7$ for cosine similarity and lexical overlap respectively), and mixed-effects models yielded consistent estimates in

the expected direction. Participants' self-report suggests that when questions had higher personal relevance, participants took more ownership (e.g. P2: "I prefer to write on my own when possible, especially when it is about something that matters to me...") but allowed the LLM to lead the responses more for less personal generative questions or when the answers were "outside of themselves" (P9). Together, these results suggest potential differences in LLM use behavior by question type.

5 Discussion

5.1 Anecdotal Questions More Immune to LLM Usage

Our preliminary findings suggest that responses to anecdotal questions show lower overlap with LLM output than for generative questions. Some participants also reported that when questions felt more personally relevant, they were more willing to take ownership of their responses. Together, these patterns suggest that anecdotal questions may be comparatively more immune to LLM delegation than generative questions. Future work can more closely examine how the interactions between anecdotal question types and the level of personalness impact LLM usage, particularly when anecdotal questions require levels of personal disclosure or recall that may exceed the participants' comfort levels. Additionally, investigating chatbots that retain personal context or interaction history may clarify whether more personalized LLM systems enable participants to craft more representative responses.

5.2 Designing Survey-Embedded LLM to Scaffold Human-Directed Responses

Our pilot study adds to prior work suggesting that online study participants use LLM to help them answer survey questions. Our preliminary results suggest that higher LLM usage was associated with questions with higher perceived cognitive effort. These findings point to opportunities for designing AI chatbots that support participants in reframing or breaking down complex questions to make them easier to answer. For example, instead of having the LLM simply respond to a question about workplace struggles with a fictional story, an embedded LLM can guide the participant to think about specific aspects of the workplace (e.g., coworkers or assignments) that might cue them to recall an instance from their own personal experience. Our participants' tendency to answer without the assistance of LLMs reflects a desire to preserve agency and authenticity when crafting responses, rather than a blanket desire to satisfice by using LLMs. This selective and deliberate use of LLMs opens up opportunities to design better human-LLM collaborative survey systems that preserve participants' sense of agency while ensuring that LLM-inclusive workflows are accounted for.

5.3 Considering LLM Usage as Data Source

Our preliminary results suggest that some participants interacted with the AI chatbot over multiple turns to craft the response that they integrated into the question response. This suggests a potential direction for downstream analysis. Rather than focusing solely on the submitted response, researchers could examine the entire

LLM interaction, including prompts, revisions, and discarded outputs, as an additional source of insight into participants' reasoning, preferences, and collaborative expectations. Reworded prompts may reflect how participants attempt to steer the LLM towards certain ideas, while discarded outputs may be regarded as non-representative statements about the participant. Considering LLMs interactions as a part of the pipeline between researcher question and final response could support more nuanced thematic and sentiment analyses from human-LLM hybrid responses.

6 Limitations

Our small sample size limits statistical power and the stability of both GEE and mixed-effects estimates; accordingly, many findings should be interpreted as descriptive trends rather than strong inferential results. Perceived effort and personalness were measured via self-report after responding and may have been confounded by chatbot use itself.

Our study's chatbot setup was able to capture real LLM usage but it may have come at the expense of ecological validity. The presence of a chatbot may have primed or induced more usage than typical. We designed the chatbot to remain hidden for each question until the participant clicked a button to use it. In this early work, capturing more instances of LLM usage can be more informative. Future work can explore what level of interface friction best mimics the LLM setup participants typically use.

Additionally, our chatbot did not retain personal history or contextual information across questions, which may differ from the LLMs participants use in everyday settings. We made this design choice intentionally to prevent context information from carrying over between survey questions. The lack of personalization may have influenced participants' decision to use the chatbot and how they interacted with it.

7 Conclusion

In a pilot study where we captured how participants used an AI chatbot to help them complete an online survey, we observed that when the AI chatbot is used, participants use initial prompts very similar to the question text and their question responses are very similar to the LLM output. Questions with higher perceived effort were associated with increased odds of using LLM assistance, suggesting a need to support the cognitive demands on the participant, perhaps through AI chatbot interactions. Participants were less likely to directly use LLM-generated content for anecdotal questions, suggesting that either LLM-generated responses were inadequate or participants wanted to retain agency in sharing their personal experiences. In an age when LLM usage is prevalent, researchers can leverage how participants prompt and modify AI chatbot responses to be a source of data to better understand the participant.

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